

NETSETOS

Production-ready AI engineering, built in India

GenAI Engineering Roadmap 2026

v1.0 • MAY 2026

14 modules. 93 lessons. Built for Indian software engineers who want to ship production GenAI systems.

INSIDE THIS ROADMAP

- Why GenAI Engineering is more than Prompt Engineering
- The 14-module learning path, module by module
- Day-1 tooling stack (Python, Colab, GCP, HuggingFace)
- The Multilingual Token Tax — Hindi costs 2-3x more
- 30-day self-study plan (free path)
- India-specific concerns (DPDP Act, Bhashini, IndicCorp)
- Recommended resources & references
- When to self-study vs. join the cohort (honest take)

Authored by Sart & Tanu • Founders, Netsetos

Hyderabad, India • netsetos.com

Join our WhatsApp Community for the daily AI brief

CONTENTS

What's in this roadmap

01	Why GenAI Engineering is more than Prompt Engineering	3
02	The 14-Module Roadmap	4
03	Day-1 Tooling Stack	8
04	The Multilingual Token Tax	10
05	30-Day Self-Study Plan (Free Path)	11
06	India-Specific Production Concerns	13
07	Recommended Resources	14
08	Cohort vs. Self-Study — When to Pay	16

How to use this roadmap. Read it once cover-to-cover (45 mins). Then pick the section that matches where you are: stuck on tooling? Section 3. Looking for a self-study path? Section 5. Wondering if the paid cohort is worth it? Section 8 has our honest take.

SECTION 01

Why GenAI Engineering ≠ Prompt Engineering

Most online GenAI content stops at: *'here's a clever prompt.'* That's a starting point, not a profession. GenAI Engineering is what happens after the demo works — when you have to ship a system that handles real users, real data, and real failure modes.

Here's the gap nobody talks about. A working ChatGPT prompt does not guarantee a working production system. Production GenAI systems fail for reasons prompt engineering never covers: rate limits, latency budgets, fallback chains, evaluation pipelines, prompt injection, hallucination detection, cost spikes from token bloat, multilingual edge cases, and compliance with Indian regulations like the DPDP Act 2023.

The five layers of real GenAI work

Layer	What it covers	Where most courses stop
1. Prompt	Crafting inputs to get useful outputs	Covered everywhere
2. Pipeline	RAG, agents, tool-calling, chains	Covered shallowly
3. Evaluation	How do you know the output is good?	Skipped
4. Production	Scale, latency, cost, fallbacks, monitoring	Skipped
5. Compliance	PII, DPDP Act, hallucination liability	Skipped

This roadmap (and the cohort it points to) covers all five layers, with India-specific context layered into every module — multilingual token costs, IndicCorp datasets, hallucination rates in Indian legal and medical use cases, GCP Mumbai vs. Yotta Shakti vs. E2E Networks deployment trade-offs, and DPDP-compliant data handling patterns.

The honest test. If you can already deploy a RAG system that handles 1,000 concurrent users, evaluates its own outputs, falls back gracefully when OpenAI rate-limits you, and stays compliant with the DPDP Act — you don't need this roadmap. You're already a GenAI engineer. If any of that sounds intimidating, keep reading.

SECTION 02

The 14-Module Roadmap

What follows is the structure of our flagship cohort — but use it as a roadmap regardless of whether you join. Each module lists what you'll build, why it matters in production, and the India-specific angle that most international courses miss.

MODULE 1**Foundations: Python for AI Engineers**

What you'll build: A typed, tested Python codebase with async I/O, dataclasses, and structured logging — the foundation every later module assumes.

Why it matters: Most production GenAI failures aren't model failures, they're Python failures. Type errors, race conditions, untested code paths.

India angle: Indian engineers often come from Java/.NET backgrounds — this module bridges that gap with idioms that match production Python.

MODULE 2**LLM Fundamentals: How Models Actually Work**

What you'll build: Token-level intuition: tokenization, embeddings, attention, sampling. You'll implement a tiny GPT from scratch (~200 lines).

Why it matters: You can't debug what you don't understand. Cost spikes, weird outputs, and context limits all trace back to fundamentals.

India angle: Hindi/Tamil/Telugu tokenize 2-3× larger than English — knowing why is the difference between a ₹5K and ₹15K monthly OpenAI bill.

MODULE 3**Prompt Engineering, Properly**

What you'll build: Structured prompts, few-shot patterns, chain-of-thought, self-consistency, and prompt versioning — treated like code, not magic.

Why it matters: Prompts in production need version control, rollback, and A/B testing. Prompt drift breaks systems silently.

India angle: Indic-language prompts often need different patterns than English — directness, formality, and code-switching matter.

MODULE 4**Embeddings & Vector Search**

What you'll build: Build a semantic search engine over your own documents using sentence-transformers, FAISS, and pgvector.

Why it matters: Vector search is the substrate under every RAG system, every recommendation engine, and most agentic memory.

India angle: Bhashini's Indic embeddings outperform multilingual-e5 on Hindi/Tamil/Telugu retrieval — we benchmark both.

MODULE 5**Retrieval-Augmented Generation (RAG)**

What you'll build: A production RAG pipeline with hybrid retrieval (BM25 + vector), reranking, and chunk-level citations.

Why it matters: Every enterprise GenAI use case is some flavor of RAG. Customer support, legal research, internal Q&A.

India angle: Indian legal RAG (over IPC sections, case law) and medical RAG (over MCI guidelines) have unique citation requirements.

MODULE 6**Advanced RAG: GraphRAG, HyDE, Self-RAG**

What you'll build: When vanilla RAG fails — multi-hop questions, domain hierarchies, sparse documents — these are the patterns that fix it.

Why it matters: Most RAG demos pass simple Q&A. Real users ask multi-step questions that vanilla retrieval can't handle.

India angle: Indian enterprise data is often hierarchical (state→district→ward) — GraphRAG fits this structure naturally.

MODULE 7**Agents, Tools, and Function Calling**

What you'll build: Build a ReAct agent that uses tools (search, calculator, SQL) with proper error handling and step caps.

Why it matters: Agents are how GenAI moves from 'answer questions' to 'do work.' But naive agents loop forever or burn tokens.

India angle: Building agents that integrate with Indian APIs (UPI, DigiLocker, Aadhaar e-KYC) requires extra care around PII.

MODULE 8**Evaluation: How Do You Know It Works?**

What you'll build: Build an eval pipeline using LLM-as-judge, human eval rubrics, and regression testing on real production traces.

Why it matters: If you can't measure your system, you can't improve it. Most GenAI shipping is held back here.

India angle: Indian-language evals are scarce — we cover how to build your own using IndicCorp and human annotators.

MODULE 9**Fine-tuning & LoRA**

What you'll build: Fine-tune a base model with LoRA on a custom dataset. Compare against prompting and RAG.

Why it matters: Fine-tuning is the right tool sometimes — but most teams reach for it before exhausting cheaper options.

India angle: When does fine-tuning beat prompting for Indic languages? We benchmark across Hindi, Tamil, Bengali.

MODULE 10**Production Deployment: GCP, Yotta, E2E**

What you'll build: Deploy a GenAI service to production with autoscaling, monitoring, and rate limiting on real Indian cloud providers.

Why it matters: Where you deploy matters: latency to your users, cost in INR, compliance with data localization.

India angle: Side-by-side cost analysis: GCP Mumbai vs. Yotta Shakti vs. E2E Networks for inference workloads, in INR/hour.

MODULE 11**Cost & Latency Optimization**

What you'll build: Caching, batching, distillation, prompt compression, and request routing — turn ₹100K/mo bills into ₹15K/mo bills.

Why it matters: GenAI cost spirals are real. We've seen startups burn 6 months of runway because nobody set up rate limits.

India angle: Yotta Shakti's GPU pricing in INR makes some workloads dramatically cheaper than US-based deployment.

MODULE 12**Safety, Guardrails, and Hallucination Detection**

What you'll build: Build PII detection (with Indian identifier patterns — Aadhaar, PAN, UPI), prompt injection guards, and output validators.

Why it matters: One leaked Aadhaar in your logs is a compliance disaster. Defaults don't catch Indian PII patterns.

India angle: Hallucination rates on Indian legal and medical questions are documented to be higher than for US contexts — guardrails matter more here.

MODULE 13**Compliance: DPDP Act, AI Liability, Data Residency**

What you'll build: Map a real GenAI system against DPDP Act 2023 requirements. Understand consent, data minimization, and the right to erasure for AI training data.

Why it matters: DPDP is now law. Penalties go up to ₹250 crore. Building DPDP-aware from day 1 is much cheaper than retrofitting.

India angle: How does DPDP apply when your model was trained outside India? When user data flows through OpenAI? We work through real cases.

MODULE 14**Capstone: Ship a Production GenAI Product**

What you'll build: End-to-end build: pick a real problem, ship it, instrument it, eval it, present it to a panel of working GenAI engineers.

Why it matters: The portfolio piece that gets you hired or funded. Companies want to see your code, not your certificate.

India angle: Capstone projects align with Indian use cases (regional language customer support, Indian legal Q&A, multilingual content moderation) so they're relevant to Indian employers.

SECTION 03

Day-1 Tooling Stack

What to install before Module 1 begins. Don't overthink this — the goal is to get unblocked, not to build the perfect environment.

Local development

- **Python 3.11+** — Don't use 3.9 for new projects. Async syntax and type-hint improvements matter.
- **uv (or pip + venv)** — Modern Python package manager. 10-100× faster than pip. Skip Anaconda.
- **VS Code or Cursor** — Cursor is VS Code with AI built-in. Free tier handles most cohort work.
- **Git + GitHub account** — Every cohort assignment is submitted as a public repo. No exceptions.

Cloud compute (free tiers first)

- **Google Colab Free** — T4 GPU, 12GB RAM, 12-hour sessions. Enough for Modules 1-7.
- **Modal Labs** — Serverless GPU, ₹/sec billing. Free tier: 30 GPU-hours/month.
- **Kaggle Notebooks** — Free P100 GPU, 30 hrs/week. Better than Colab for some training jobs.

LLM APIs (you'll need at least one)

- **OpenAI API** — Industry standard. Add ₹500 credit to start. Set hard spending limits.
- **Anthropic Claude API** — Best for long context and code. ₹500 credit recommended.
- **Together.ai or Groq** — Open-source models at API speed. Cheaper than OpenAI for many tasks.
- **Bhashini APIs** — Free for Indic-language NLP tasks, hosted by Gol.

Data & vector stores

- **HuggingFace account** — Free model hub, datasets, spaces. Required for Modules 4 onwards.
- **Pinecone Free Tier** — 1M vectors free. Easiest managed vector DB.
- **Postgres + pgvector** — Self-hosted vector search. Production-grade, free.
- **Weaviate or Qdrant** — Open-source alternatives. Worth knowing for cohort capstone.

Production cloud (Module 10+)

- **GCP free tier** — ₹25K credit for new accounts. Mumbai region for low India latency.

- **Yotta Shakti** — Indian sovereign GPU cloud. Cheaper for sustained workloads.
- **E2E Networks** — Indian cloud, INR billing, A100 access from ~₹140/hour.

Don't buy anything yet. Every tool listed has a free tier sufficient for the first half of the roadmap. Spending money before Module 5 is a sign of procrastination, not progress.

SECTION 04

The Multilingual Token Tax

This section alone might save you ₹50,000 a year.

Most LLM tokenizers were trained predominantly on English text. When you feed them Hindi, Tamil, Telugu, Bengali, or any other Indic language, they tokenize *much less efficiently* — meaning you pay 2-3× more for the same semantic content. This is the multilingual token tax, and almost every Indian GenAI startup discovers it the hard way after the first month's bill.

How big is the tax in practice?

Sentence	English tokens	Hindi tokens	Cost multiplier
"How do I reset my password?"	7	—	1.0×
"मैं अपना पासवर्ड कैसे रीसेट करूँ?"	—	21	3.0×
"What is the weather today?"	6	—	1.0×
"आज मौसम कैसा है?"	—	16	2.7×
"Tell me about Indian history."	6	—	1.0×
"मुझे भारतीय इतिहास के बारे में बताओ।"	—	23	3.8×

Three ways to reduce the tax

1. Use models with better Indic tokenizers. Sarvam-1, OpenHathi, Krutrim, and Anthropic Claude tokenize Indic text more efficiently than older OpenAI models. Benchmark before committing to a vendor.

2. Translate strategically. For non-customer-facing operations (RAG retrieval, classification, search), translate Indic input to English, run the LLM, translate output back. The translation is cheap; the LLM call is expensive.

3. Cache and compress aggressively. Indic queries hit the cache layer harder than English ones because you're paying 3× for cache misses. Set up semantic caching from day 1.

Real example from a cohort student. A Bengaluru fintech was running customer support in Hindi through GPT-4. Monthly bill: ₹2.4 lakh. After switching to a hybrid (translate to English, GPT-4o-mini, translate back) the bill dropped to ₹38,000. Same response quality. That's the tax in real money.

SECTION 05

30-Day Self-Study Plan (Free Path)

If you're not ready to join the paid cohort — or want to test whether GenAI engineering is for you before committing — here's a 30-day path using only free resources. About 1.5 hours/day.

Week 1 — Fundamentals (~10 hours)

- **Day 1-2:** Andrej Karpathy's 'Let's build GPT' video (2 hrs). Watch, don't just skim.
- **Day 3-4:** Tokenization deep-dive — Karpathy's tokenizer video + run the OpenAI tokenizer tool on Hindi text.
- **Day 5-6:** 3Blue1Brown's neural network series (the visual intuition matters).
- **Day 7:** Build a tiny embedding-based search over your own notes (50 lines of Python).

Week 2 — Prompt Engineering & APIs (~10 hours)

- **Day 8-9:** Anthropic's prompt engineering course (free, 2 hrs).
- **Day 10-11:** Get an OpenAI API key. Add ₹500 credit. Run 50+ prompts on different tasks.
- **Day 12-13:** DeepLearning.AI 'ChatGPT Prompt Engineering for Developers' (free, 1.5 hrs).
- **Day 14:** Ship a working ChatBot using the OpenAI API + Streamlit. Deploy free on HuggingFace Spaces.

Week 3 — RAG & Agents (~12 hours)

- **Day 15-16:** LangChain RAG tutorial (the 'Build a RAG application' walkthrough).
- **Day 17-18:** Build a RAG app over a real Indic document set (e.g., IPC sections from indiankanoon.org).
- **Day 19-20:** ReAct agent paper + LangGraph tutorial. Build an agent that uses 2+ tools.
- **Day 21:** Add evaluation — log every query, score retrieval quality, find your top 5 failures.

Week 4 — Production & Capstone (~12 hours)

- **Day 22-23:** Hamel Husain's 'AI Evals' blog series (hamel.dev/blog/posts/evals-faq).
- **Day 24-25:** Deploy your RAG app to a real cloud — GCP free tier or Modal.
- **Day 26-27:** Add monitoring, rate limiting, and a fallback chain.
- **Day 28-30:** Write a public blog post about what you built. Share on LinkedIn. Tag us — we'll amplify it.

If you finish this plan and want more depth. The cohort goes 3-4× deeper on every topic, with India-specific labs, weekly live sessions, and 1:1 code review. The free path will tell you whether GenAI engineering is for you. The cohort will get you production-ready faster.

SECTION 06

India-Specific Production Concerns

DPDP Act 2023 — what every GenAI engineer needs to know

The Digital Personal Data Protection Act, 2023 is now in force. For GenAI systems, the most consequential parts are: **(1)** consent must be specific, free, and revocable; **(2)** data principals have a right to erasure — including from your training data; **(3)** penalties scale up to ₹250 crore per violation. If your system trains on or stores user data, you need a DPDP-aware architecture, not a retrofit.

Indian PII patterns most libraries miss

Off-the-shelf PII detectors (Microsoft Presidio's defaults, OpenAI's content filter) catch SSNs, emails, and US phone numbers. They miss **Aadhaar** (12-digit, often spaced as XXXX-XXXX-XXXX), **PAN** (10-character alphanumeric), **UPI VPAs** (handle@bank), **vehicle registration numbers**, and **Indian phone formats**. Module 12 builds custom regex + entity recognition for these.

The hallucination problem is worse for Indian context

GPT-4 and Claude have higher hallucination rates on Indian legal questions (IPC sections, case citations) and Indian medical guidelines than on US equivalents — because their training data is thinner here. For any application touching Indian law or medicine, RAG with authoritative source citations is non-optional.

Data residency: when does India force you to keep data here?

Sectoral rules (RBI for fintech, IRDAI for insurance, MeitY for e-gov) increasingly require certain data categories to remain on Indian soil. This affects which LLM providers and cloud regions you can use. GCP Mumbai, AWS Mumbai/Hyderabad, Yotta Shakti, and E2E Networks are the safe defaults; OpenAI's ChatGPT Enterprise has India data residency, but the standard API does not.

Indic NLP resources every Indian GenAI engineer should know

- **Bhashini** — Government of India's NLP platform. Free Indic translation, ASR, TTS APIs.
- **IndicCorp v2** — 20.9 billion tokens across 24 Indic languages, AI4Bharat.
- **Samanantar** — Largest publicly available parallel corpus for 11 Indic languages.
- **IndicTrans2 / IndicBERT** — Open Indic translation and embedding models from AI4Bharat.
- **Sarvam-1, OpenHathi, Krutrim** — Indic-tuned LLMs. Worth benchmarking against OpenAI for Indic tasks.

SECTION 07

Recommended Resources

These are the resources we recommend to cohort students and rely on ourselves. Free unless noted.

Foundational courses (free)

- **Andrej Karpathy** — 'Neural Networks: Zero to Hero' YouTube series. Mandatory.
- **Stanford CS336: Language Modeling from Scratch** — full lecture videos online.
- **HuggingFace LLM Course** — practical, hands-on, free, regularly updated.
- **DeepLearning.AI short courses** — Andrew Ng's 1-2 hour focused tutorials.
- **fast.ai** — Jeremy Howard's 'Practical Deep Learning for Coders.'

Books (worth buying)

- **'Build a Large Language Model from Scratch'** — Sebastian Raschka. The single best book on LLM internals.
- **'Designing Machine Learning Systems'** — Chip Huyen. Production ML, not just GenAI, but the principles transfer.
- **'AI Engineering'** — Chip Huyen, 2024. The companion book to this roadmap, frankly.

Newsletters & blogs (subscribe)

- **Hamel Husain's blog** (hamel.dev) — best practical GenAI eval content on the internet.
- **Sebastian Raschka's Ahead of AI** — weekly ML/LLM research distilled.
- **The Batch** — DeepLearning.AI's weekly AI roundup.
- **Latent Space** — Swyx's podcast + newsletter, especially good on agent architectures.

Code repositories (read & fork)

- **Karpathy/nanoGPT** — minimal GPT implementation, ~300 lines.
- **rasbt/LLMs-from-scratch** — Sebastian Raschka's book code.
- **langchain-ai/langchain** — vast, but the cookbook examples are gold.
- **AI4Bharat repos** — IndicTrans2, IndicBERT, Samanantar.

Indian creators worth following

- **Krish Naik** — YouTube, 1.3M+ subscribers. Hindi + English data science content.
- **Nitish Singh (CampusX)** — Hinglish-first ML/DL teaching.
- **Dhaval Patel (Codebasics)** — practical data analytics + AI.
- **Build Fast with AI** — Satvik Paramkusham, IIT Delhi alum, agent-focused content.

And, of course

- **Netsetos YouTube** (youtube.com/c/nETSETOSTECH) — 500+ free tutorials, GenAI series ongoing.
- **Netsetos WhatsApp Community** — daily AI brief from Sart, doubt-solving with Tanu.
- **netsetos.com** — Cohort enrollment + course catalog.

SECTION 08

Cohort vs. Self-Study — When to Pay

We're not going to pretend the cohort is for everyone. Most people who download this PDF should do the free 30-day plan first. Some should join the cohort. Here's how to tell which group you're in.

Stay free if...

- You're still figuring out whether GenAI is the right direction for your career.
- You have 10+ hours/week for self-directed learning and stay accountable to yourself.
- Your immediate goal is to learn, not to ship a production system in < 6 months.
- ₹15,000–₹30,000 is genuinely tight right now — this is real money, don't pretend it isn't.
- You're already a senior engineer who can read papers and figure out tooling unaided.

Join the cohort if...

- You have a specific GenAI thing you need to ship in < 6 months and want a guided path.
- You've tried self-study before and stalled — you know structure helps you finish.
- You want India-specific context (DPDP, Indic languages, Indian cloud) that no global course covers.
 - You want feedback on your code from working GenAI engineers, not just internet strangers.
- You want a peer group of Indian engineers building similar things to learn alongside.

What the cohort actually includes

- **14 modules, 93 lessons** — every topic in this roadmap, taught with India-specific labs.
- **Live weekly sessions** — Sart and Tanu, real-time Q&A, recorded for replay.
- **Code review** — every assignment reviewed by working GenAI engineers, not bots.
- **Capstone project** — ship something real, presented to a panel that includes hiring managers.
- **Private cohort Discord** — peer group of Indian engineers building similar things.
- **Lifetime access** — to recordings, code, and future cohort updates.
- **Cohort 1 launch pricing** — significantly below market for the first batch.

Our honest commitment. If you join Cohort 1 and complete every assignment in the first two modules and decide it's not for you, we refund 100%. No questions, no friction. We'd rather have 100 students who are sure than 200 who aren't.

READY TO BUILD?

Cohort 1 enrollment opens soon.

14 modules. 93 lessons. Live cohort with Sart & Tanu.
India-first context. Production-grade builds. Real code, not slides.

YOUTUBE / youtube.com/c/nETSETOSTECH • **WEB** / netsetos.com

Drop questions in our WhatsApp Community — we answer daily.

About the authors

Sart (Sarthak Kumar) — Co-founder of Netsetos. 15+ years in software engineering across Python, Django, GCP, and now GenAI systems. Builder of the Netsetos YouTube channel (youtube.com/c/nETSETOSTECH, 500+ tutorials, 24K+ subscribers). Currently building the GenAI Engineering cohort and the Daily AI Brief in our WhatsApp Community.

Tanu Varshney — Co-founder of Netsetos. Curriculum designer, technical author, and the primary voice on Netsetos's LinkedIn presence. Leads cohort operations, doubt-solving, and the alumni community.

Netsetos Technologies LLP — registered in Hyderabad, India (LLPIN AAW-7974). EdTech focused on production-grade software and AI engineering education for Indian engineers.